

The `-vv` represents the verbosity in the Ansible output. You can use up to four 'v's for a more detailed output. The `-K` option prompts for the `sudo` password for the guest CentOS user account.

## The PostgreSQL master

The PostgreSQL configuration files need to be updated on the master node with various settings required for replication. The database server is restarted and a replication user is also created. The `replica_password` can be encrypted and stored using Ansible Vault, when used in production. The playbook to set up the PostgreSQL master instance is as follows:

```
- name: Setup pgmaster
  hosts: pgmaster
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [pgmaster]

  vars:
    replica_password: 'replica123'

  tasks:
    - name: Update /var/lib/pgsql/9.6/data/postgresql.conf
      lineinfile:
        path: /var/lib/pgsql/9.6/data/postgresql.conf
        regexp: "{{ item.regexp }}"
        line: "{{ item.line }}"
        with_items:
          - { regexp: "#listen_addresses = 'localhost'", line: "listen_addresses = '{{ ansible_default_ipv4.address }}'" }
          - { regexp: '#wal_level = minimal', line: 'wal_level = hot_standby' }
          - { regexp: '#synchronous_commit = on', line: 'synchronous_commit = local' }
          - { regexp: '#archive_mode = off', line: 'archive_mode = on' }
          - { regexp: "#archive_command = ''", line: "archive_command = 'cp %p /var/lib/pgsql/9.6/archive/%f'" }
          - { regexp: '#max_wal_senders = 0', line: 'max_wal_senders = 2' }
          - { regexp: '#wal_keep_segments = 0', line: 'wal_keep_segments = 2' }
          - { regexp: "#synchronous_standby_names = ''", line: "synchronous_standby_names = 'slave01'" }

    - name: Create archive directory
      file:
        path: /var/lib/pgsql/9.6/archive
        mode: 0700
        owner: postgres
        group: postgres
        state: directory
```

```
- name: Update pg_hba.conf
  blockinfile:
    path: /var/lib/pgsql/9.6/data/pg_hba.conf
    insertafter: '#host replication postgres
::1/128 ident'
    block: |
      # Localhost
      host replication replica
      127.0.0.1/32 md5

      # PostgreSQL Master IP address
      host replication replica
      {{ ansible_default_ipv4.address }}/32 md5

      # PostgreSQL Slave IP address
      host replication replica
      {{ hostvars['host2'].ansible_default_ipv4.address }}/32
      md5

- name: Restart PostgreSQL server
  systemd:
    name: postgresql-9.6
    enabled: yes
    state: restarted

- name: Create replication user
  shell: sudo -u postgres psql -c "CREATE USER replica
REPLICATION LOGIN ENCRYPTED PASSWORD '{{ replica_password
}}'"
```

The above playbook invocation and sample output is shown below, for reference:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/postgresql.yml --tags pgmaster -K
SUDO password:

PLAY [Common pre-requisites] *****

TASK [Gathering Facts] *****
ok: [host2]
ok: [host1]

PLAY [Setup pgmaster] *****

TASK [Gathering Facts] *****
ok: [host1]

TASK [Update /var/lib/pgsql/9.6/data/postgresql.conf] *****

changed: [host1] => (item={'regexp': "#listen_
addresses = 'localhost'", 'line': "listen_addresses =
'192.168.122.174'"})
changed: [host1] => (item={'regexp': '#wal_level = minimal',
```